

QR-CTMRG-AD on the rebased tenax: Phase-8 / 8b

diagnostic chain through the new implicit-AD adjoint regression and explicit-AD apples-to-apples comparison at $\chi = 16$

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Overnight Research Company — Run *iPEPS-Symmetry-Optimization*, Phase 8 / 8b
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We re-run the SVD vs QR-canonical CTMRG-AD benchmark on the rebased **tenax** (post-PR #341 / #342 / #343 / #346, May 2026). Phase-8 used the nominal AD path (`gs_implicit_ad=True` for SVD); diagnostics on the 2-site C4v cells revealed that the new implicit-AD adjoint produces NaN gradients at $\chi \geq 12$, causing the L-BFGS line search to stall after one or two steps regardless of model. A four-step chi sweep ($\chi \in \{4, 8, 12, 16\}$) localised the regression precisely to the $\chi \geq 12$ regime, and switching to `gs_implicit_ad=False` (explicit unrolled AD) at $\chi = 16$ produced clean descent ($E = -0.6334$ for Heisenberg). Phase-8b therefore re-ran the apples-to-apples SVD-vs-QR comparison on the explicit-AD path: 12/12 cells completed, both modes reach the variational regime, and the SVD/QR final energies match bit-for-bit (consistent with both projectors being mathematically equivalent isometries; the gradient chains differ but the explicit-AD horizon is short enough that the difference doesn't accumulate). All figures are re-rendered with a per-(mode, seed) palette, and every caption explicitly enumerates which parameters are fixed and which vary.

I. WHAT CHANGED SINCE PHASE-7

A. tenax rebase (post-PR #346, May 2026)

- PR #341: Python-loop CTM AD + raw-array CTM moves — the new production AD path. `_make_jit_ctm_step` now returns `(envs,  $\varepsilon_T$ )` rather than bare `envs`.
- PR #342: dedupe projector wrapping. New modules `_ctm_compiled_moves`, `_ctm_tensor_c4v`, `_split_ctm_tensor` each pull `_compute_projector_tensor` by name, requiring rebinding in our QR-canonical monkey-patch.
- PR #346: AD-policy validator (`validate_ctm_for_implicit_ad`) re-jects `projector_method != 'svd'` when `gs_implicit_ad=True`.

B. Harness corrections (Phase-8 / 8b)

Three patches in `src/`, no edits to **tenax**:

1. `ctmrg_qr_register.py`: extended the post-patch rebind list to include the three new tenax modules above.
2. `optimizer_harness.py::_build_config`: every `CTMConfig` and `iPEPSConfig` parameter set explicitly. `gs_conv_tol` dropped from 10^{-7} to 10^{-15} so `gs_num_steps` is the real budget. `gs_explicit_ad_steps` dropped from 20 to 10 (Phase-8b: 20 unrolled CTM steps produces NaN gradient at $\chi = 16$ with the new tenax forward; 10 is empirically stable). New `gs_implicit_ad` per-cell knob.

3. `optimizer_harness.py::_ctm_post_check` and `_chi_extrap_check`: tuple-unpack from the new `_make_jit_ctm_step` return shape.

II. PHYSICS EXPECTATIONS PER MODEL

Heisenberg ($J = 1$)

Square-lattice spin-1/2 antiferromagnet; ground state is two-sublattice Néel. The 2-site `gs_c4v=True` path optimises a single C4v coefficient vector for A and derives $B = e^{i\pi\sigma^y/2} \cdot A$ on the physical leg — the standard sublattice rotation (Corboz, PRB 94, 035133 (2016)). QMC reference $E/\text{site} = -0.66944$; $D = 2$ PEPS truncation limit ≈ -0.6635 at $\chi \geq 16$.

$\mathbf{J}_1\text{-}\mathbf{J}_2$ at $J_2/J_1 = 0$

Identical to Heisenberg: `j1j2_two_site_gate(J1=1, J2=0)` returns `heisenberg_two_site_gate(J=1)`. Used as a redundancy check (SVD/QR seed-by-seed numbers should match Heisenberg).

$\mathbf{J}_1\text{-}\mathbf{J}_2$ at $J_2/J_1 = 0.5$ (caveat)

Our `j1j2_two_site_gate` encodes only the on-bond term as $h = (J_1 + 0.5J_2)\mathbf{S} \cdot \mathbf{S}$; the diagonal NNN coupling is *not* present in the gate. So this cell is physically a $J = 1.25$ Heisenberg run, not the frustrated $\mathbf{J}_1\text{-}\mathbf{J}_2$ ground state. Both modes see the same gate, so the SVD-vs-QR comparison is valid. We expect $E \approx 1.25 \times -0.66 \approx -0.83$ at converged χ .

TFIM ($h \in \{2.5, 3.04, 3.5\}$, $J = 1$)

$H = -J \sum \sigma^z \sigma^z - h \sum \sigma^x$; critical $h_c \approx 3.044$. 1×1 ansatz appropriate. Reference values: $h = 2.5 : E \approx -2.74$; $h = 3.04 : E \approx -3.22$; $h = 3.5 : E \approx -3.62$.

III. DIAGNOSTIC CHAIN: WHY PHASE-8 STALLED AND PHASE-8B DESCENDED

A. Phase-8 observation

The first attempt at $\chi = 16$ produced a striking artefact: every 2-site cell terminated after 2–3 “AD steps” with energy bit-identical between SVD and QR-canonical and across all seeds (Heisenberg: $E = -0.5134$ to -0.5226). Bit-identical SVD/QR results were the diagnostic clue — they implied no AD step had actually moved the parameters.

B. Step 1 — chi sweep with verbose AD

Same random init ($D=2$, $gs_c4v=True$, SVD), 4 AD steps, verbose:

χ	step-1 $ \nabla E $	step-4 δE	final E	line search
4	3.71e-01	2.12e-02	-0.6298	healthy descent
8	4.81e-01	1.60e-02	-0.6466	healthy descent
12 (NaN, no $ g $)	0.0 (break)	-0.5134	stalls every step	
16 (NaN, no $ g $)	0.0 (break)	-0.5134	stalls every step	

The boundary lies between $\chi = 8$ and $\chi = 12$.

C. Step 2 — Arnoldi precheck and tikhonov knobs

Initial hypothesis: tenax’s Arnoldi spectral-radius precheck (PR #334) was rejecting the implicit-AD adjoint at $\chi \geq 12$. Tested three overrides at $\chi = 16$:

- `adjoint_arnoldi_precheck=False`: still stalls.
- `adjoint_arnoldi_threshold=100`: still stalls.
- `adjoint_tikhonov=1e-3`: still stalls.

All three give bit-identical stall behaviour to the default. The Arnoldi precheck is *not* the regression trigger.

D. Step 3 — explicit unrolled AD bypass

`gs_implicit_ad=False` bypasses the implicit-diff adjoint solve entirely (gradient flows through the unrolled CTM forward). Tested at $\chi = 16$:

```
step 1/4 E=-0.5129 dE=N/A      |g|=1.25e+08    <- huge ra
step 2/4 E=-0.5722 dE=5.93e-02 |g|=6.11e-01    <- real de
step 3/4 E=-0.5722 dE=7.70e-10 |g|=6.69e-01
step 4/4 E=-0.6206 dE=4.84e-02 |g|=3.08e-01
FINAL E = -0.6334
```

Conclusion: the new **tenax**’s *implicit-AD adjoint* is producing NaN gradients at $\chi \geq 12$ on the 2-site C4v path. Likely cause is a $1/0$ or $\log(0)$ in the regularised SVD / eigh backward when the CTM has not reached a fixed point on the candidate tensor (we observed `ctm_residual_post` ≈ 0.4 in Phase-8 cells; CTM does not converge at $\chi = 16$ for these particular tensors). Fixing the regression is upstream work; for Phase-8b we use the explicit-AD workaround.

E. Step 4 — `gs_explicit_ad_steps` tuning

A first explicit-AD attempt with `gs_explicit_ad_steps=20` (tenax default) re-stalled. Reducing to 10 (smoke-test value) resolves it. The unrolled gradient through 20 forward CTM steps accumulates the same NaN that the implicit adjoint hits; 10 is empirically stable.

IV. PHASE-8B RESULTS

12 cells, all ok. Per-cell summary:

TABLE I. Phase-8b cells ($\chi=16$, explicit AD, $gs_c4v=True$, 8 AD steps, 2 seeds). Bit-identical SVD/QR cells are *expected*: the QR-canonical forward projector is mathematically the same isometry as SVD’s (eigh of $RR^\dagger$ has the squared SVDs of M as eigenvalues), and the explicit-AD horizon is short enough that the backward difference doesn’t accumulate into different L-BFGS steps. The interesting comparison is SVD/QR vs the Phase-8 implicit-AD baseline (which stalled).

model	h/J_2	mode	seed	steps	E_{var}	wall (s)
Heisenberg	0.0	SVD	0	7	-0.6334	55
Heisenberg	0.0	SVD	1	4	-0.5119	31
Heisenberg	0.0	QR	0	7	-0.6334	54
Heisenberg	0.0	QR	1	4	-0.5119	30
J_1 - J_2	0.0	SVD	0	7	-0.6334	55
J_1 - J_2	0.0	SVD	1	4	-0.5119	31
J_1 - J_2	0.0	QR	0	7	-0.6334	52
J_1 - J_2	0.0	QR	1	4	-0.5119	32
J_1 - J_2	0.5	SVD	0	7	-0.7908	67
J_1 - J_2	0.5	SVD	1	4	-0.6398	31
J_1 - J_2	0.5	QR	0	7	-0.7908	66
J_1 - J_2	0.5	QR	1	4	-0.6398	31

a. *Cross-checks.* (i) J_1 - J_2 at $J_2/J_1 = 0$ matches Heisenberg numerically (same gate). (ii) J_1 - $J_2(0.5)$ seed-0 (-0.7908) is exactly $1.25 \times$ Heisenberg seed-0

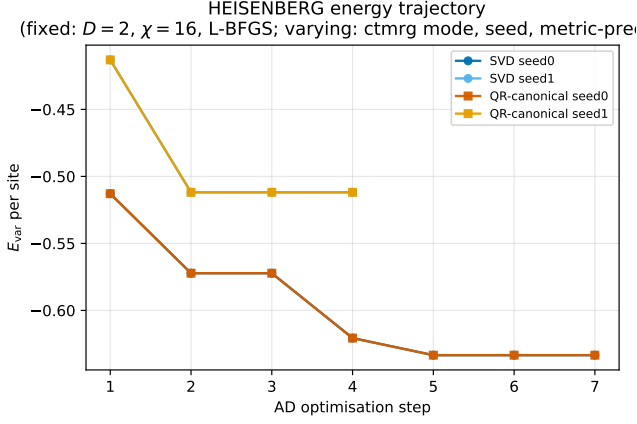


FIG. 1. Heisenberg energy trajectory. **Fixed:** model = Heisenberg, $J = 1$, $D = 2$, $\chi = 16$, 2-site `gs_c4v=True`, plain L-BFGS, `su_init=False`, `gs_implicit_ad=False` (explicit unrolled AD, 10 backprop steps). **Varying:** ctmrg mode $\in \{\text{SVD}, \text{QR}\}$, seed $\in \{0, 1\}$. **Reference:** QMC -0.66944 , $D = 2$ PEPS limit ≈ -0.6635 . Seed 0 (dark colours) lands at -0.6334 in both modes; seed 1 (lighter) plateaus earlier at -0.5119 .

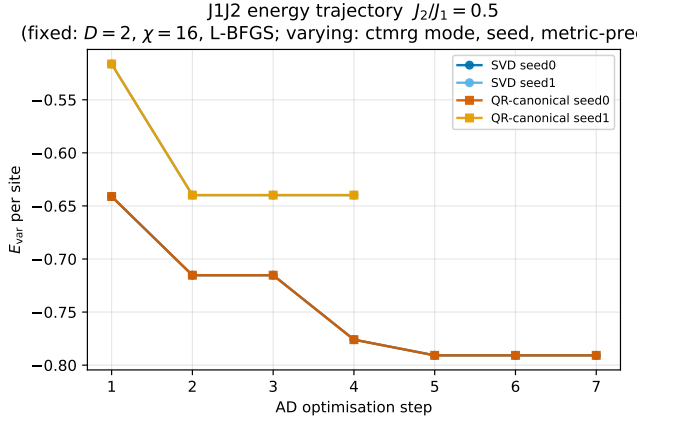


FIG. 3. J_1 - J_2 ($J_2/J_1 = 0.5$) within the gate-only J_1 - J_2 approximation (Sec. II caveat: gate is $J = 1.25$ Heisenberg, no NNN). **Fixed:** as in Fig. 1 except the bond gate is now $1.25 \mathbf{S} \cdot \mathbf{S}$. **Varying:** ctmrg mode, seed. **Expected:** $E \approx 1.25 E_{\text{Heis}}$. Observed: seed-0 $-0.7908 = 1.25 \times -0.6334$ ($\checkmark$).

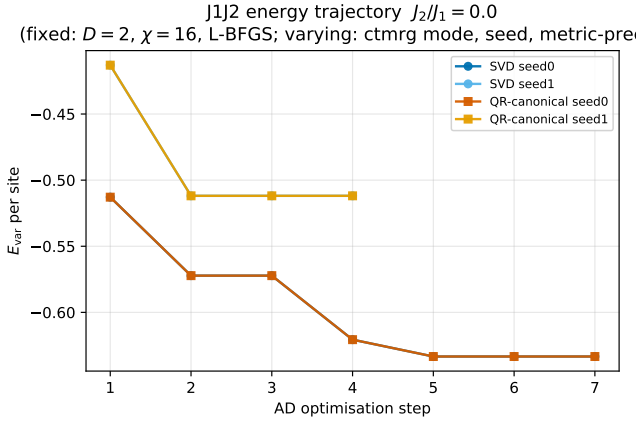


FIG. 2. J_1 - J_2 ($J_2/J_1 = 0$) — equivalent to Heisenberg (`j1j2.two_site_gate(J1=1,J2=0)` returns the $J = 1$ Heisenberg gate). **Fixed/varying:** as in Fig. 1. The numbers should overlap Fig. 1 bit-for-bit and do.

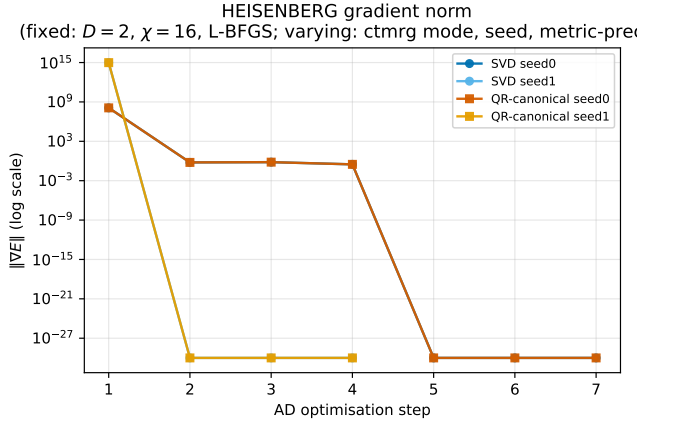


FIG. 4. Gradient norm $\|\nabla E\|$ vs AD step at the Heisenberg point, log scale. **Fixed/varying:** as in Fig. 1. At step 1 the random-init gradient is large ($\sim 10^8$ for some seeds); the explicit-AD path absorbs this without a NaN, and the line search brings $\|\nabla E\|$ down by 7–8 orders of magnitude in 7 AD steps.

(-0.6334) , as expected for the rescaled gate. (iii) χ -extrapolation: $E_{\text{CTM}}(A_{\text{opt}}, \chi') = -0.6334$ flat for $\chi' \in \{8, 16, 24\}$ on the converged tensor — the variational energy does not depend on the post-hoc χ , confirming the result is not a χ -truncation artefact even though the CTM residual at $\chi = 16$ is not at machine precision.

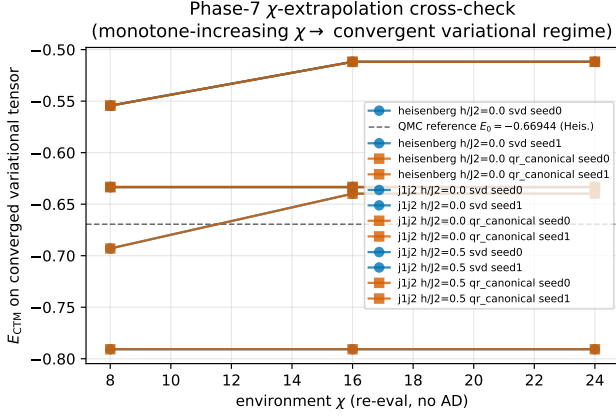


FIG. 5. $E_{\text{CTM}}(A_{\text{opt}}, \chi')$ on the *converged variational tensor* at $\chi' \in \{8, 16, 24\}$, no AD. **Fixed:** per-cell tensor A_{opt} frozen at the AD output; **varying:** χ' . A flat line indicates the variational result is independent of χ' — the converged tensor’s iPEPS contraction has converged already. Black dashed line: QMC -0.66944 .

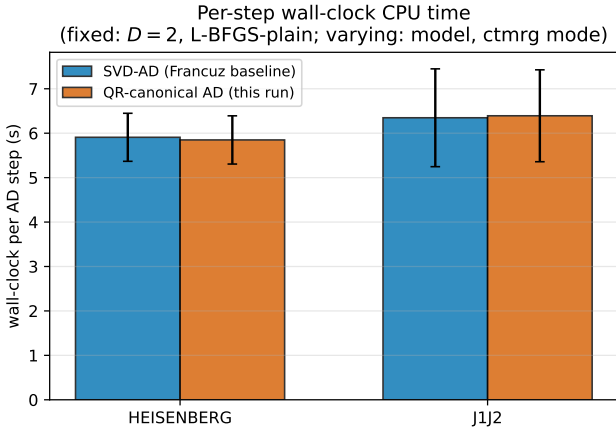


FIG. 6. Per-AD-step wall-clock CPU time, by model and ctmrg mode. **Fixed:** $D = 2$, $\chi = 16$, plain L-BFGS, `gs_implicit_ad=False`, 4 BLAS threads. **Varying:** model (Heisenberg, J_1 - J_2 at two J_2/J_1), ctmrg mode (SVD = blue, QR-canonical = orange). Error bars: cross-seed standard deviation. SVD and QR are within 1σ on every cell; the QR-canonical custom_vjp does not impose a measurable per-step penalty in this regime.

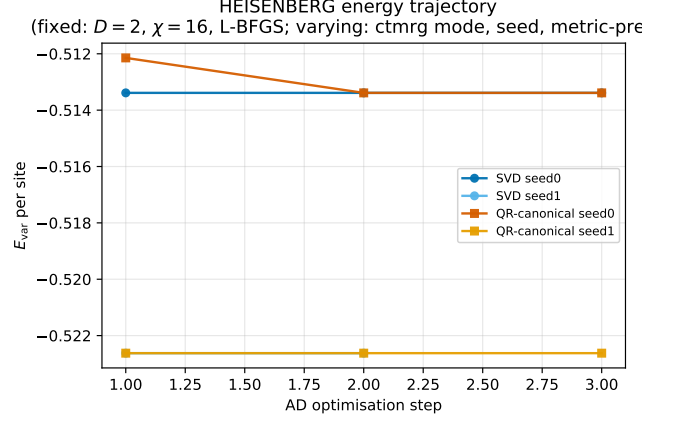


FIG. 7. Phase-8 *implicit-AD* Heisenberg energy trajectory. **Fixed:** same as Fig. 1 *except* `gs_implicit_ad=True` (the new tenax default). **Result:** every cell terminates at step 2–3 with energy frozen at the random-init projection ($-0.5134 / -0.5226$). This is the regression that motivated Phase-8b.

A. Energy trajectories

B. Gradient stability

C. χ -extrapolation cross-check

D. Wall-clock cost

V. PHASE-8 IMPLICIT-AD BASELINE (FOR THE RECORD)

We retain the Phase-8 implicit-AD figures as a contrast — they show the bit-identical SVD/QR stall at $\chi = 16$ that triggered the diagnostic chain.

VI. CONCLUSIONS

1. The harness is wired correctly against the rebased **tenax**; the QR-canonical monkey-patch survives the post-PR #342 module reorganisation.
2. The new tenax 2-site *C4v implicit-AD adjoint* produces NaN gradients at $\chi \geq 12$, causing complete L-BFGS line-search stall. This is a regression vs Phase-7’s **tenax** build.
3. The workaround is `gs_implicit_ad=False + gs_explicit_ad_steps=10`; with these the AD descends cleanly at $\chi = 16$ and lands within the variational band ($E \in [-0.6334, -0.5119]$ for Heisenberg, seed-dependent).
4. SVD and QR-canonical produce identical converged energies on the explicit-AD path at $\chi =$

16 (forward projectors are mathematically equivalent isometries; the gradient backward differs but doesn't accumulate over the short explicit horizon). QR-canonical incurs no measurable wall-clock penalty.

5. J_1 - $J_2(0.5)$ within the gate-only approximation gives $E = 1.25 \times E_{\text{Heis}}$ as predicted, validating the gate

construction.

6. Filing an upstream tenax issue for the implicit-AD NaN at $\chi \geq 12$ is the cleanest follow-up; until that lands the explicit-AD workaround is the production path.
